Paper I – Basic Sciences in Urology (MCh Urology, Summer 2015)

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I-1. Describe in details basic patho-physiology of urine production and transport

Type: LAQ (25 marks)

Introduction

- Urine is formed by filtration, reabsorption, secretion and concentration, then propelled to a low-pressure bladder.

A. Functional anatomy

- Glomerular barrier (fenestrated endothelium, basement membrane, podocytes) restricts by size and negative charge, so albumin is excluded.
- **Juxtamedullary nephrons** (long loops) build the medullary gradient.

B. Renal blood flow (RBF) and autoregulation

- RBF ~25% of cardiac output. The medulla is hypoxic (10–20 mmHg), endangering the thick ascending limb and S3 segment.
- **Autoregulation** keeps RBF and GFR (glomerular filtration rate) constant at mean arterial pressure 80–180 mmHg:
  - **Myogenic:** the afferent arteriole constricts when stretched and dilates (prostaglandins) when pressure falls; NSAIDs (non-steroidal anti-inflammatory drugs) abolish this.
  - **Tubuloglomerular feedback:** low NaCl at the macula densa releases renin; angiotensin II constricts the efferent arteriole. Angiotensin blockade abolishes this.

C. Glomerular filtration

- GFR ~125 mL/min (180 L/day); filtration fraction ~20%.
- **Net filtration pressure** ≈ glomerular hydrostatic 60 – Bowman's space 18 – oncotic 32 = **~10 mmHg**.
- Efferent constriction lowers RBF but **raises** GFR; afferent constriction lowers both. Obstruction raises Bowman's space pressure, so GFR falls.
- **Also know:** GFR estimated from creatinine or cystatin C (race-free 2021 equations).

D. Tubular handling

Segment	Key transporter	Handling	Drug/disease link
Proximal tubule	Na^+/H^+ exchanger, SGLT2 (sodium–glucose cotransporter 2)	~**65%** Na^+ and water (iso-osmotic); all glucose; most bicarbonate and phosphate; secretes drugs	SGLT2 inhibitors
Thin descending limb	Aquaporin-1	Water out, solute stays: fluid concentrates	–
Thick ascending limb	NKCC2 (Na^+-K^+-$2Cl^-$)	~**25%** NaCl; water-impermeable "diluting segment"	Loop diuretics
Distal convoluted tubule	NCC (Na^+-Cl^- cotransporter)	5–7% Na^+; Ca^{2+} (parathyroid hormone)	Thiazides lower urinary Ca^{2+}
Collecting duct	ENaC (epithelial Na^+ channel), AQP2	Principal cells: Na^+/K^+ (aldosterone), water (ADH); intercalated cells: acid	Amiloride, spironolactone; distal RTA (renal tubular acidosis)

E. Concentration

- **Countercurrent multiplication:** ascending limbs absorb NaCl without water, raising medullary osmolality from 300 to ~**1200 mOsm/kg**; urea recycling adds about half. Vasa recta exchange without washing out the gradient.
- **Antidiuretic hormone (ADH)** is released by high osmolality or low volume. Signal chain: V2 receptor → cAMP (cyclic adenosine monophosphate) → protein kinase A → aquaporin-2 (AQP2) inserted into the apical membrane → water reabsorbed. Concentration needs both a hypertonic medulla and AQP2; urine osmolality ranges ~50–1200 mOsm/kg.

Hormone	Site	Effect
Angiotensin II, aldosterone	Efferent arteriole, proximal tubule; collecting duct	Na^+ retention, K^+/H^+ loss, GFR maintained
Atrial natriuretic peptide	Afferent arteriole, collecting duct	Natriuresis, lowers renin
Parathyroid hormone	Proximal and distal tubule	Phosphaturia, Ca^{2+} retention

F. Ureteric transport

- **Pacemaker:** atypical smooth muscle cells at the minor calyces/pelvicalyceal border.
- Peristalsis is **myogenic**: the excised or denervated ureter keeps contracting. Nerves (T11–L1, S2–4) carry only pain.
- Calcium entry triggers action potentials spread by gap junctions; calcium–calmodulin then activates myosin light-chain kinase.
- Walls coapt behind each bolus: 2–6 waves/min at 20–80 cmH_2O. Very high flow makes boluses coalesce, dilating the ureter **without obstruction**.
- Ureterovesical tunnel length:diameter 5:1 prevents reflux; bladder pressure >40 cmH_2O stops efflux.

Factor	Effect on transport
Obstruction	Acute: stronger contractions; chronic: dilatation, lost coaptation
Infection	Gram-negative endotoxin inhibits peristalsis (dilatation without obstruction)
Pregnancy	Right > left dilatation: compression at pelvic brim, progesterone

Factor	Effect on transport
Drugs	α1-blockers and calcium-channel blockers relax (medical expulsive therapy); NSAIDs lower pelvic pressure in colic

G. Bladder storage and voiding

- **Storage** below ~10 cmH_2O: compliance, sympathetic β3 detrusor relaxation, α1 bladder-neck and pudendal sphincter tone.
- **Voiding:** pelvic parasympathetic (S2–4) acetylcholine on M3 receptors contracts the detrusor; nitric oxide relaxes the urethra.

H. Pathophysiology of obstruction (unilateral: triphasic)

```
Phase 1 (0–1.5 h): RBF up,   pressure up    (afferent dilatation:
                                             prostaglandins, nitric oxide)
Phase 2 (1.5–5 h): RBF down, pressure up    (efferent constriction)
Phase 3 (>5 h):    RBF down, pressure down  (afferent constriction: angiotensin II,
                                             thromboxane A2, endothelin)
```

- Ischaemic injury starts **within 2 hours** of complete block.
- **Tubules:** down-regulated aquaporins and Na^+ transporters cause concentrating defect, salt wasting and distal RTA.
- **Fibrosis** (final path to failure): stretch, inflammation, TGF-β (transforming growth factor-β), angiotensin II, microRNAs.
- **Bilateral obstruction or solitary kidney:** pressure stays high longer and natriuretic peptides and urea accumulate, so **post-obstructive diuresis** follows relief in ~50% (vs ~2% unilateral).
- **Clinical:** hypertension (~77% bilateral, ~20% unilateral); contralateral hypertrophy; pyelorenal backflow acts as a pop-off; forniceal rupture causes urinoma.
- **Also know:** immature kidneys fail to develop; outlet obstruction causes detrusor hypertrophy then decompensation; diuretic MAG3 (mercaptoacetyltriglycine) renogram (half-time >40 min = obstructed); Whitaker test (>15–20 cmH_2O = obstructed); tolvaptan in polycystic disease.

Diagram to draw: Nephron with transporters; triphasic graph.

> **Key points to remember:**
> - RBF 25% of cardiac output; GFR 125 mL/min; net filtration pressure ~10 mmHg.
> - Proximal 65%, loop 25%, distal 5–7% of Na^+; ADH → V2 → AQP2.
> - Myogenic peristalsis from calyceal pacemaker; post-obstructive diuresis 2% vs 50%.

I-2. Describe female pelvic anatomy in relation to prolapse and SUI

Type: LAQ (25 marks)

Introduction

- Support comes from the **levator ani, endopelvic fascia and ligaments, perineal membrane and bony pelvis**. Failure causes **pelvic organ prolapse (POP)** and **stress urinary incontinence (SUI)**.

A. Bony pelvis

- **Bony landmarks** and dangers: **sacrospinous ligament** (pudendal bundle), **Cooper's ligament** (bleeding), **sacral promontory** (left common iliac vein, right ureter).

B. Pelvic floor muscles

Muscle	Attachment	Function
Pubococcygeus	Pubis to vagina, perineal body, anus (coccygeus is separate)	Closes hiatus, lifts urethra
Puborectalis	Sling behind anorectal junction	Anorectal angle
Iliococcygeus	Arcus tendineus levator ani to coccyx	Horizontal **levator plate**

- Levator tone closes the **urogenital hiatus**, through which all POP passes.
- **Muscle fails first, fascia second:** once injury opens the hiatus, ligaments carry the load and fail.
- Levator defects in up to **20% of primiparous women**; avulsion predicts advanced prolapse.

C. Endopelvic fascia and ligaments

- **Cardinal and uterosacral ligaments** suspend the apex (ureter close to uterosacral at the cervix).
- Pubocervical fascia attaches laterally to the **arcus tendineus fasciae pelvis (ATFP, "white line")**, pubis to just above the ischial spine.
- **Rectovaginal fascia** supports the posterior wall; **pubourethral ligaments** anchor the mid-urethra.

D. DeLancey's levels of support (1992)

Level	Region	Structures	Defect causes
I – suspension	Cervix/upper third	Cardinal–uterosacral complex	Uterine/vault prolapse, enterocele
II – attachment	Middle third	ATFP, levator fascia	Cystocele, rectocele
III – fusion	Distal third	Perineal membrane, perineal body	Urethral hypermobility, perineal descent

- **Central cystocele:** midline weakness, rugae lost. **Paravaginal cystocele:** ATFP detachment (Richardson 1981), rugae preserved.
- Level I is the keystone: always correct the apex.

E. Urethra and continence

- Continence zone: **proximal and mid-urethra**; striated sphincter (compressor urethrae, urethrovaginal sphincter) plus oestrogen-sensitive submucosa (~30% of closure).

Theory	Mechanism	Implication
Pressure transmission (Enhorning 1961)	Hypermobile urethra leaves the abdominal pressure zone	Restore position (Burch)
Hammock (DeLancey 1994)	Urethra lies on anterior vaginal wall and endopelvic fascia attached to ATFP and levator; abdominal pressure compresses it against this layer	Restore backing
Integral theory (Petros & Ulmsten 1990)	Mid-urethral pubourethral ligament laxity	Mid-urethral sling
Intrinsic sphincter deficiency (ISD) (McGuire 1978)	Urethral function fails	Sling, bulking

- ISD: Valsalva leak point pressure <60 or maximal urethral closure pressure <20 cmH_2O. Most SUI has some ISD; Burch does poorly in severe ISD.

F. Innervation

- **Pudendal nerve (S2–4):** sphincters and perineum; through Alcock's canal near the ischial spine; stretched in delivery.
- **Nerve to levator ani (S3–5):** on the muscle's pelvic surface, separate from pudendal; neuropathy in about a quarter of primiparous women.
- Autonomic pelvic splanchnic and hypogastric nerves: injured in radical pelvic surgery.

G. Risk factors (Bump & Norton)

- Predisposing: genetics, collagen disorders. Inciting: **vaginal delivery** (strongest; forceps worse), hysterectomy. Promoting: obesity, constipation, cough. Decompensating: ageing.

H. POP-Q (Pelvic Organ Prolapse Quantification, Bump 1996)

- Nine points in cm relative to the **hymen (0)**: inside negative, outside positive; measured on maximal strain, recorded on a 3 × 3 grid.

Point	Definition
Aa / Ap	Anterior / posterior wall 3 cm inside hymen (−3 to +3)
Ba / Bp	Most distal point of anterior / posterior wall (−3 if no prolapse)
C	Cervix or vault cuff
D	Posterior fornix (omitted after hysterectomy)
gh, pb, tvl	Genital hiatus, perineal body, total vaginal length (at rest)

Stage	Most distal point
0	No prolapse
I	>1 cm above hymen
II	Within ±1 cm of hymen
III	>1 cm below hymen, < (tvl – 2) cm
IV	Complete eversion (≥ tvl – 2)

- **Also know:** Baden–Walker grades 0–4; stage I–II often normal (POSST study).

I. Examination

- Lithotomy and **standing** with Valsalva; Sims speculum for each compartment; lateral sulci (paravaginal defect); rectovaginal examination.

- **Cough stress test**, repeated with prolapse reduced (**occult SUI**); **Q-tip** angle change >30° = hypermobility; levator strength and avulsion; urodynamics if ISD or mixed symptoms.

J. How anatomy guides surgery

Defect	Operation	Principle
SUI	**Mid-urethral sling**: retropubic (Ulmsten 1995, TVT [tension-free vaginal tape]) or transobturator (Delorme 2001)	Mid-urethral support
SUI	**Burch colposuspension**	Fascia to Cooper's ligament
SUI – ISD	Autologous sling at bladder neck, bulking	Compression
Paravaginal / central cystocele	Paravaginal repair / anterior colporrhaphy	Reattach to ATFP / midline plication
Apical	**Sacrocolpopexy** (most durable)	Mesh to ligament over S1
Apical (vaginal)	**Sacrospinous fixation** 1.5–3 cm medial to spine; uterosacral suspension (cystoscopy for ureters)	Avoids pudendal bundle
Rectocele	Posterior or site-specific repair, perineorrhaphy	Levels II–III
Elderly, not sexually active	Colpocleisis	Obliteration

- **April 2019:** the US FDA (Food and Drug Administration) stopped transvaginal mesh for anterior POP; slings unaffected.

Diagram to draw: DeLancey levels I–III; hammock cross-section.

> **Key points to remember:**
> - DeLancey: I suspension, II attachment (ATFP), III fusion; level I is the keystone.
> - Enhorning 1961, DeLancey 1994, Petros & Ulmsten 1990, McGuire ISD 1978.
> - Central cystocele rugae lost, paravaginal preserved; POP-Q II = ±1 cm of hymen.

I-3a. Newer generation CT scan

Type: SAQ (10 marks)

Introduction

- Computed tomography (CT) is the reference imaging for stones, masses and trauma. Attenuation is expressed in Hounsfield units (HU): water 0, air −1000.

Evolution

Generation	Feature
Helical (~1989)	Continuous table movement, volumetric data
Multidetector (1998 → 320-row)	Sub-millimetre slices, 3D reconstruction
Dual-source/dual-energy (2005)	Two energies: material analysis
Photon-counting (US approval 2021)	Energy-sorts each photon: less noise, dose and metal artefact

Dual-energy CT (DECT)

- Two energies (e.g. 80 and 140 kVp) compared pixel by pixel: **uric acid vs non-uric-acid stones**, **virtual unenhanced** images (one phase fewer) and **iodine maps** (hyperdense cyst vs enhancing mass).

Low-dose CT

Protocol	Dose
Conventional abdomen/pelvis	10–20 mSv
Low-dose	<3–4 mSv
Ultra-low-dose	1–2 mSv

- Stone sensitivity stays ~95%; use standard dose if body mass index >30. EAU (European Association of Urology): ultrasound first; AUA (American Urological Association): CT first.

Reconstruction

- Filtered back projection is noisy; iterative reconstruction halves dose; **deep learning reconstruction** gives good images at ~1 mSv.

Contrast phases and protocols

Phase	Timing	Use
Arterial	20–30 s	Renal arteries, donor anatomy, active bleeding
Corticomedullary	30–70 s	Hypervascular (clear cell) tumours, renal vein
Nephrographic	90–120 s	Most sensitive for renal masses
Excretory	7–10 min	Collecting system, ureter, urine leak

- **CT urography:** non-contrast, nephrographic and excretory phases (~20–30 mSv). **Split-bolus** merges phases, lowering dose but may miss subtle urothelial lesions.
- **Renal mass:** enhancement ≥20 HU = solid, 10–20 equivocal, <10 cyst; Bosniak 2019 for cysts.
- **Adrenal:** <10 HU unenhanced = lipid-rich adenoma; otherwise 15-minute washout (absolute >60% or relative >40% = adenoma).
- **Also know:** stone HU (uric acid 200–400, calcium oxalate 600–1200; indinavir invisible); bone window for fragments; partial volume causes pseudoenhancement of small cysts;

ALARA (as low as reasonably achievable) dosing, since radiation risk has no threshold.

Diagram to draw: Timeline: axial → helical → multidetector → dual-source → photon-counting.

Key points to remember:

- Doses: conventional 10–20, low-dose <3–4, ultra-low 1–2 mSv.
- DECT: uric acid stones, virtual unenhanced, iodine maps.
- Enhancement ≥20 HU = solid; deep learning enables ~1 mSv CT.

I-3b. History of Laparoscopy

Type: SAQ (10 marks)

Introduction

- Laparoscopy is endoscopic viewing of the peritoneal cavity through a pneumoperitoneum; it grew from optics to robotic surgery.

Optics and access

Year	Pioneer	Milestone
1806	**Bozzini**	**Lichtleiter**, candle-lit light conductor
1877	Nitze	Distal electric light and lenses
1901	**Kelling**	Insufflated abdomen of a **dog**
1910	**Jacobaeus**	First human series; "laparoscopy"
1938	**Veress**	Spring-loaded needle
1966	**Hopkins**	**Rod-lens system**
1970s–80s	**Semm**	Automatic insufflator; operative laparoscopy
1970s	Hasson	Open access
1987	Mouret	Video cholecystectomy

- **Also know:** Desormeaux 1853, first therapeutic endoscopy; fibreoptics (Tyndall's principle) for flexible scopes; CO_2 became the standard insufflant.

Urological milestones

Year	Pioneer	Milestone
1976	**Cortesi**	Impalpable testis, first urological use
1990–91	**Clayman**	**First laparoscopic nephrectomy**; nephroureterectomy 1991
1992	**Gaur (India)**	**Balloon retroperitoneal access**
1995	**Ratner**	Live-donor nephrectomy
1997	Schuessler	Laparoscopic radical prostatectomy (slow, not adopted)
2000	Guillonneau & Vallancien	Standardised prostatectomy
2000	**da Vinci**; **Binder**	Robot cleared in US; first robotic prostatectomy
2000s	**Menon**	Vattikuti Institute Prostatectomy
2007	Raman; Rane	Single-port nephrectomy

Why the robot spread

- Laparoscopic prostatectomy stalled because intracorporeal suturing was hard. The robot gave wristed instruments and a 3D magnified view.
- da Vinci generations: S 2006 (fourth arm), Si 2009 (dual console), Xi 2014 (flexible docking), SP 2018 (single port).

Indian contribution and recent advances

- Gaur's balloon made retroperitoneoscopy practical; indigenous robot SSI Mantra with telesurgery.
- New robots (Versius, Hugo, Hinotori), telesurgery, haptic feedback.

Diagram to draw: Timeline 1806 → 1901 → 1910 → 1938 → 1990 → 2000.

Key points to remember:

- Bozzini 1806; Kelling 1901 (dog); Jacobaeus 1910 (human); Veress 1938; Hopkins rod-lens.
- Urology: Cortesi 1976; Clayman 1990; Gaur 1992; Ratner 1995.
- da Vinci 2000; Binder first robotic prostatectomy; robot solved suturing barrier.

I-3c. Renin-angiotensin system

Type: SAQ (10 marks)

Definition

- Hormonal cascade controlling **blood pressure, extracellular volume, Na^+/K^+ balance and GFR (glomerular filtration rate)**.

Cascade

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Angiotensinogen (liver)
   ↓ renin (juxtaglomerular cells) — rate-limiting
Angiotensin I
   ↓ ACE (angiotensin-converting enzyme), mainly lung
Angiotensin II → AT1 / AT2 receptors
   ↓ ACE2
Angiotensin (1-7) → Mas receptor (protective)
```

Juxtaglomerular apparatus and renin release

- Granular cells in the afferent arteriole secrete renin and act as baroreceptors; the **macula densa** senses tubular NaCl.

Renin stimulated by	Renin inhibited by
Low renal perfusion pressure (baroreceptor)	High perfusion pressure
Low NaCl at macula densa	High NaCl at macula densa
Sympathetic β1 activity	Angiotensin II (negative feedback)
Prostaglandins, hypovolaemia	Atrial natriuretic peptide, β-blockers, NSAIDs (non-steroidal anti-inflammatory drugs)

Actions of angiotensin II

- **AT1 receptor:** vasoconstriction; **aldosterone** (distal Na^+ retention, K^+ loss); direct proximal Na^+ reabsorption; thirst and ADH (antidiuretic hormone); **fibrosis**.
- **AT2 receptor:** vasodilatation; fetal urinary tract development.
- **Intrarenal:** efferent constriction maintains GFR when perfusion falls, so ACE inhibitors and ARBs (angiotensin receptor blockers) drop GFR in bilateral or solitary-kidney renal artery stenosis.

Urological relevance

- **Renovascular hypertension (Goldblatt kidney):** atherosclerosis (older men, ostial) or fibromuscular dysplasia (young women). Diagnose by duplex ultrasound, computed tomography or magnetic resonance angiography, captopril renography. **Renal vein renin ratio >1.5** with contralateral suppression predicts cure by revascularisation or nephrectomy.
- **Page kidney:** subcapsular haematoma compresses parenchyma; ACE inhibitor/ARB, then drainage.
- **Reninoma (juxtaglomerular cell tumour):** young women, ≤3 cm mass, refractory hypertension, **hypokalaemia**; excision cures most.
- **Obstruction:** hypertension (~77% bilateral, ~20% unilateral); angiotensin II drives fibrosis.

- **Also know:** post-traumatic hypertension (arteriovenous fistula, scarring); reflux nephropathy (renin-guided segmental resection); hyperkalaemia with angiotensin blockade in CKD (chronic kidney disease).
- **Drugs:** ACE inhibitors (cough, angio-oedema from bradykinin), ARBs, aliskiren (direct renin inhibitor), aldosterone antagonists; all contraindicated in pregnancy.

Diagram to draw: Juxtaglomerular apparatus and cascade.

Recent advances

- Renal denervation (US approval 2023) for resistant hypertension.

> **Key points to remember:**
>
> - Renin is rate-limiting; released by low pressure, low NaCl, β1.
> - Efferent constriction maintains GFR, so ACE inhibitors risk renal failure in bilateral stenosis.
> - Renin ratio >1.5 lateralises; reninoma excision cures most.

I-3d. Patho-physiology of Brain stem death

Type: SAQ (10 marks)

Definition

- **Irreversible loss of the capacity for consciousness and to breathe** from permanent brainstem failure; legally death. Donation after brain death (DBD) keeps circulation until procurement, minimising warm ischaemia.

Pathophysiology

```
Brain injury → oedema → raised intracranial pressure (ICP)
 → falling cerebral perfusion → herniation ("coning")
 → Pons: Cushing reflex (hypertension, bradycardia)
 → Medulla: catecholamine storm (hypertension, arrhythmia)
 → Loss of sympathetic tone → vasodilatation, hypotension
```

- The storm injures myocardium and causes pulmonary oedema. Loss of antidiuretic hormone causes **diabetes insipidus (DI)**: polyuria, hypernatraemia; thyroid hormone and cortisol fall.
- **Hypothermia** (hypothalamic loss) and **disseminated intravascular coagulation (DIC)** from brain thromboplastin.

Diagnosis

- **Preconditions:** apnoeic coma of known irreversible cause; exclude sedatives, relaxants, hypothermia (35–36 °C by protocol), metabolic causes.
- **Brainstem reflexes, all must be absent:**

Test	How it is done	Cranial nerves
Pupillary light reflex	Pupils fixed, no response to bright light	II, III
Corneal reflex	Touch cornea; no blink	V, VII
Oculovestibular (caloric)	~50 mL ice-cold water into each ear (drum visible); no eye movement	VIII, III, VI
Oculocephalic (doll's eye)	Turn head; eyes move with the head	VIII, III, VI
Motor response	Central painful stimulus (supraorbital pressure); no facial grimace	V, VII
Gag and cough	Stimulate pharynx; pass suction catheter down trachea; no response	IX, X

- **Apnoea test:** pre-oxygenate with 100% oxygen, disconnect the ventilator with oxygen insufflation, and watch for breathing effort until **arterial CO_2 ($PaCO_2$) ≥60 mmHg**. No effort = positive. Abort if hypotension, desaturation or arrhythmia.
- **Ancillary tests** (electroencephalography (EEG), cerebral angiography, perfusion scan, transcranial Doppler) are not mandatory; used when clinical testing cannot be completed.

Indian law

- **THOA (Transplantation of Human Organs Act) 1994**, amended 2011, Rules 2014: **4-member board** (hospital in-charge, independent specialist, neurologist/neurosurgeon, treating doctor; no transplant team member); **two tests at least 6 hours apart**.

Donor management

- **Rule of 100s:** systolic pressure >100 mmHg, urine >100 mL/h, PaO_2 (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L.
- **Vasopressin** preferred (also treats DI); thyroid hormone, methylprednisolone, insulin.
- **Also know:** cold storage at 4 °C (University of Wisconsin solution); donor exclusions (metastatic cancer, melanoma, untreatable viral infection).

Recent advances

- Hypothermic machine perfusion reduces delayed graft function.

Diagram to draw: Herniation sequence leading to storm, DI, hypothermia, DIC.

> **Key points to remember:**
>
> - Raised ICP → Cushing reflex → storm → hypotension, DI, hypothermia, DIC.
> - Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
> - THOA: 4-member board, two tests ≥6 h apart; Rule of 100s.

I-3e. Male reproductive immunology

Type: SAQ (10 marks)

Introduction

- Haploid germ cells appear at puberty, after self-tolerance is established, so their antigens are "foreign". The testis is therefore **immune-privileged**.

Immune privilege

Anatomical barrier: blood–testis barrier (BTB)

- **Tight junctions between Sertoli cells**, formed at puberty, divide a **basal compartment** (spermatogonia, plasma-like fluid) from an **adluminal compartment** (later spermatocytes, spermatids).

Functional barrier

- A small antigen leak at the **rete testis, efferent ductules and epididymis** induces suppressor T cells and tolerance; a large load provokes immunity.
- Sertoli and local mediators: galectin-1, TGF-β (transforming growth factor-β), interferon-γ, soluble Fc receptor, **androgens**.
- Insult before puberty does not induce antibodies; the barrier also limits chemotherapy (isolated testicular relapse).

Antisperm antibodies (ASA)

- **Causes:** obstruction (**vasectomy** commonest), torsion, trauma, biopsy, infection, heat (varicocele, cryptorchidism), genetic (human leukocyte antigen HLA-B28).
- **Effects:** agglutination, poor motility, failure to cross cervical mucus and bind the zona; head-bound antibody matters most.
- **Indications to test:** isolated asthenozoospermia, agglutination, unexplained infertility.

Test	Method
Mixed antiglobulin reaction (MAR)	Fresh semen + immunoglobulin (Ig) G/A-coated beads + antiglobulin; beads stick to antibody-coated motile sperm
Immunobead test	Washed sperm + beads coated with anti-human IgG/IgA; gives % and site (head/tail) of binding

- **≥50% of motile sperm bound** is significant (WHO [World Health Organization] 2010).

Treatment	Comment
Corticosteroids	Trials conflict; risk of avascular necrosis; not recommended
Sperm washing + intrauterine insemination	Bypasses cervical mucus; modest benefit (bound antibody remains)
IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)	Preferred; fertilisation largely unaffected

Infection and inflammation

- **Leukocytospermia:** $>1 \times 10^6$ peroxidase-positive leukocytes/mL. Reactive oxygen species and cytokines damage sperm.
- **Orchitis:** mumps (post-pubertal, atrophy); bacterial (*Chlamydia*/gonococcus if young, *E. coli* if older); tuberculous; **autoimmune**, primary (antibodies with azoospermia) or secondary to vasculitis.

- **Also know:** sperm DNA fragmentation testing; prostatitis harms sperm via oxidative stress and ASA; seminal plasma suppresses female tract immunity.

Diagram to draw: Seminiferous tubule with Sertoli tight junctions and compartments.

> **Key points to remember:**
> - BTB = Sertoli tight junctions; plus active tolerance (galectin-1, TGF-β, androgens).
> - ASA: vasectomy commonest; ≥50% bound significant; IVF/ICSI.
> - Leukocytospermia >1 million/mL.

I-3f. Principles of antimicrobial therapy in UTI

Type: SAQ (10 marks)

Definitions

- **Uncomplicated** urinary tract infection (UTI): healthy non-pregnant women, normal tract. **Complicated:** men, pregnancy, abnormal tract, catheter, immunosuppression.
- **Recurrent:** ≥2 in 6 months or ≥3 in 12 months.

Principles

- **Site decides the level:** urine in cystitis, tissue elsewhere; nitrofurantoin and fosfomycin lack tissue levels: **cystitis only**.

Killing pattern	Drugs	Dosing goal
Time-dependent (time above MIC [minimum inhibitory concentration])	β-lactams, carbapenems	Frequent doses or extended infusion
Concentration-dependent (peak:MIC)	Aminoglycosides, fluoroquinolones (FQs)	High once-daily dose

- **Resistance thresholds:** trimethoprim–sulfamethoxazole (TMP-SMX) if local resistance ≤20%; oral FQ for pyelonephritis if ≤10%.
- **Collateral damage:** FQs and cephalosporins select resistance and *C. difficile*; avoid them in cystitis.
- Culture all but simple cystitis; **drain** obstructed infected kidneys; shortest course.

Empirical choices

Scenario	Drug	Duration
Cystitis (women)	Nitrofurantoin 100 mg twice daily; fosfomycin 3 g; pivmecillinam 400 mg; TMP-SMX 160/800 mg twice daily	5 d; single dose; 3–5 d; 3 d
Cystitis (men)	TMP-SMX or per culture	7 d (= 14 d, Drekonja 2021)
Pyelonephritis (oral)	Ciprofloxacin 500 mg twice daily; levofloxacin 750 mg daily	7 d; 5 d
Pyelonephritis (intravenous)	Ceftriaxone 1 g daily; gentamicin 5–7 mg/kg once daily	10–14 d; oral when afebrile
Complicated / catheter UTI	Aminoglycoside or 3rd-generation cephalosporin per culture; change catheter	7–14 d
Urosepsis	Broad-spectrum **within 1 h**, source control	–

Asymptomatic bacteriuria

- Treat **only in pregnancy and before mucosa-breaching urological procedures**. Not in elderly, diabetic, catheterised or spinal cord injury patients.
- **Pregnancy:** avoid FQs and TMP-SMX; cephalexin is suitable.

Prophylaxis

- Recurrent: vaginal oestrogen, **methenamine hippurate** (non-inferior to antibiotics, ALTAR 2022), cranberry, post-coital antibiotics.
- Surgical: single dose within 1 hour; ≤24 h for percutaneous nephrolithotomy and prostheses.
- **Also know:** ESBL (extended-spectrum β-lactamase) producers; nitrofurantoin avoided if creatinine clearance <30 mL/min; gepotidacin (US 2025).

Diagram to draw: Concentration–time curve showing peak:MIC and time above MIC.

Key points to remember:

- Recurrent = ≥2 in 6 months or ≥3 in 12 months.
- Cystitis: nitrofurantoin 5 d, fosfomycin single, TMP-SMX 3 d (resistance ≤20%).
- Pyelonephritis: ciprofloxacin 7 d or levofloxacin 5 d (FQ resistance ≤10%).

Rapid Revision Sheet – Paper I

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

I-1. Describe in details basic patho-physiology of urine production and transport (LAQ)

- RBF 25% of cardiac output; GFR 125 mL/min; net filtration pressure ~10 mmHg.
- Proximal 65%, loop 25%, distal 5–7% of Na^+; ADH → V2 → AQP2.
- Myogenic peristalsis from calyceal pacemaker; post-obstructive diuresis 2% vs 50%.

I-2. Describe female pelvic anatomy in relation to prolapse and SUI (LAQ)

- DeLancey: I suspension, II attachment (ATFP), III fusion; level I is the keystone.
- Enhorning 1961, DeLancey 1994, Petros & Ulmsten 1990, McGuire ISD 1978.
- Central cystocele rugae lost, paravaginal preserved; POP-Q II = ±1 cm of hymen.

I-3a. Newer generation CT scan (SAQ)

- Doses: conventional 10–20, low-dose <3–4, ultra-low 1–2 mSv.
- DECT: uric acid stones, virtual unenhanced, iodine maps.
- Enhancement ≥20 HU = solid; deep learning enables ~1 mSv CT.

I-3b. History of Laparoscopy (SAQ)

- Bozzini 1806; Kelling 1901 (dog); Jacobaeus 1910 (human); Veress 1938; Hopkins rod-lens.
- Urology: Cortesi 1976; Clayman 1990; Gaur 1992; Ratner 1995.
- da Vinci 2000; Binder first robotic prostatectomy; robot solved suturing barrier.

I-3c. Renin-angiotensin system (SAQ)

- Renin is rate-limiting; released by low pressure, low NaCl, β1.
- Efferent constriction maintains GFR, so ACE inhibitors risk renal failure in bilateral stenosis.
- Renin ratio >1.5 lateralises; reninoma excision cures most.

I-3d. Patho-physiology of Brain stem death (SAQ)

- Raised ICP → Cushing reflex → storm → hypotension, DI, hypothermia, DIC.
- Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
- THOA: 4-member board, two tests ≥6 h apart; Rule of 100s.

I-3e. Male reproductive immunology (SAQ)

- BTB = Sertoli tight junctions; plus active tolerance (galectin-1, TGF-β, androgens).
- ASA: vasectomy commonest; ≥50% bound significant; IVF/ICSI.
- Leukocytospermia >1 million/mL.

I-3f. Principles of antimicrobial therapy in UTI (SAQ)

- Recurrent = ≥2 in 6 months or ≥3 in 12 months.
- Cystitis: nitrofurantoin 5 d, fosfomycin single, TMP-SMX 3 d (resistance ≤20%).
- Pyelonephritis: ciprofloxacin 7 d or levofloxacin 5 d (FQ resistance ≤10%).